Sarvesh Prasanth Venkatesan

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EDUCATION

University of Southern California — Los Angeles, CA
M.S. in Electrical and Computer Engineering - Machine Learning — *2025-2027*

Vellore Institute of Technology — Vellore, India
B.Tech. in Electronics and Communication Engineering — *2019 - 2023*

EXPERIENCE

Dynamic Robotics and Control Laboratory | University of Southern California — Dec 2025 - Present
Graduate Researcher | Advisor: Prof. Quan Nguyen — *Los Angeles, CA*

- Designing a **demonstration-guided RL** framework for dexterous manipulation with SEA module, formulating task objectives and reward shaping from VR teleoperation demonstrations.
- Developing a **vision-conditioned hierarchical policy** for humanoid loco-manipulation, training a flow matching motion generator with RL residual actions for a phaseless pipeline and sim2real transfer across varying object configurations.

Stochastic Robotics Lab | Indian Institute of Science — Jul 2024 – Aug 2025
Project Associate | Advisor: Prof. Shishir Kolathaya — *Bengaluru, India*

- Designed a **recurrent-transformer**-based generalist quadruped policy leveraging offline imitation learning to fuse diverse locomotion skills, achieving zero-shot skill transfer across 5+ unseen quadruped morphologies. [Webpage]
- Implemented **RL locomotion** policies for 70 kg Stoch5 quadruped; reduced **sim2real gap** by via novel *Ground-Reaction Force* and *adaptive Raibert-inspired rewards*, improving push recovery by **75–98%** on *30◦ low-traction inclines*. [Videos]
- Explored and implemented **vision-based RL** policies for locomotion on challenging terrains. [Code]

Mobile Robotics Lab | Indian Institute of Science — Sep 2023 – Jun 2024
Project Associate | Advisors: Prof. Debashish Ghose — *Bengaluru, India*

- Developed a centralized **UAV-UGV multi-agent** system with coverage path planning, relaying georeferenced points of interest to the UGV for autonomous ground level inspection. [Video]
- Built onboard autonomy stacks using cuVSLAM for VIO and autonomous navigation in GPS-denied unstructured environments.
- Collaborated on **LocalizeSORT**, a multi-object tracking framework with world-coordinate association, achieving **+15% HOTA** and **+30% prediction accuracy**. [Paper]

Centre for Non-Destructive Evaluation | Indian Institute of Technology Madras — Sep 2022 – Jul 2023
Intern | Advisors: Prof. Prabhu Rajagopal — *Chennai, India*

- Developed a full software stack for a Shopping Assistance Robot integrating GMapping SLAM, AMCL localization, and ROS Nav2 for autonomous retail navigation. [Video]
- Implemented MoveIt with OMPL for motion planning and YOLO with RGBD depth projection for 3D object pose estimation, enabling end-to-end autonomous grocery pick and place.

PUBLICATIONS

- Narayanan PP, **Sarvesh Prasanth Venkatesan**, Srinivas Kantha Reddy, Shishir Kolathaya. GRoQ-LoCO: Generalist and Robot-agnostic Quadruped Locomotion Control using Offline Datasets. *arXiv.*
- Srihari Vemuru, Kumar Ankit, **Sarvesh Prasanth Venkatesan**, Debashish Ghose, Shishir Kolathaya. Yield Prediction and Counting Using World Coordinates. Under Review at *Computers and Electronics in Agriculture*.

PROJECTS

Humanoid WBC (Github)

- Implemented a scandot based visual humanoid locomotion policy for terrain adaptive walking, based on this paper. [Video]
- Experimented with CNN based depth encoders as an alternative to scandots for terrain perception.
- Fine-tuning Nvidia GR00T N1.5 on Unitree G1 for language-guided pick-and-place in IsaacSim via behavior cloning on task-specific demonstration data.

Quadruped Motion Planning (Github)

- Implemented RRT* and optimization-based planners for quadruped navigation in MuJoCo, with collision checking and cost functions accounting for quadruped foothold constraints. [Video]
- Learned locomotion skills including stair climbing, slope traversal, and dynamic walking using PPO with terrain curriculum, integrated with global motion plans for hierarchical end-to-end autonomous navigation.

SKILLS

Programming: Python, C++.
ML/Robotics: PyTorch, JAX, ROS, MoveIt, IsaacSim, IsaacLab, MuJoCo, mjlab, OpenCV, PX4,.
Software: Docker, Git, AWS, WandB, Fusion360, LaTeX, Ubuntu Linux.